

Mutual linearity from classical Markov response theory

Timur Aslyamov^{1,*} and Massimiliano Esposito^{1,†}

¹*Complex Systems and Statistical Mechanics, Department of Physics and Materials Science,
University of Luxembourg, 30 Avenue des Hauts-Fourneaux, L-4362 Esch-sur-Alzette, Luxembourg*

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This short note explains how known mutual-linearity relations for Markov jump processes follow from classical response theory. When the rates of one edge change, an exact finite-response identity shows that the changes in all stationary probabilities share a common factor. Eliminating this factor gives the relations between probabilities, observables, and currents, even far from equilibrium. We also explain how the same argument applies when one coefficient of a linear system changes and recovers the known relations in Laplace space and between stationary complex activities in chemical reaction networks.

I. INTRODUCTION

Stationary probabilities and currents generally depend nonlinearly on transition rates. Nevertheless, varying the rates of one edge produces straight-line relations between these quantities, known as mutual linearity [1–7]. The purpose of this short explanatory note is to clarify the common origin of the basic probability and current relations using the classical finite-response identity [8, 9]. Written with the Drazin inverse, this identity shows that all probability changes share one common factor. Eliminating it gives the relations between probabilities, observables, and currents. This gives a single short argument for both the stationary and the known Laplace-space relations [1, 6, 7], and explains when the same property holds in other linear systems. Applied to the complex-space Laplacian, the argument also recovers the homogeneous relations between stationary complex activities in chemical reaction networks [10].

We do not claim to cover all the results of Refs. [1–7, 10]. Their more detailed approaches also give results outside the scope of this note. These include explicit network formulas for the coefficients relating currents [1], expressions for these coefficients in terms of mean first-passage times [2], relations for several input currents [3], bounds on stationary probabilities and observables [5], and relations between Laplace transforms of first-hitting-time distributions [7].

II. EXACT STATIONARY RESPONSE

The finite-response identity shown below dates back to the 1960s [8] and was originally formulated for Markov chains. Here we simply rewrite it for continuous-time Markov jump processes on a finite state space.

We consider a finite irreducible Markov jump process, $\partial_t \mathbf{p} = \mathbf{W} \mathbf{p}$, where W_{ij} is the transition rate from j to i , and $W_{jj} = -\sum_{i \neq j} W_{ij}$. The stationary distribution satisfies $\mathbf{W} \boldsymbol{\pi} = 0$ and $\mathbf{1}^\top \boldsymbol{\pi} = 1$. For a second irreducible genera-

tor $\mathbf{W}' = \mathbf{W} + \Delta \mathbf{W}$, the classical fundamental-matrix identity takes the form

$$\Delta \boldsymbol{\pi} \equiv \boldsymbol{\pi}' - \boldsymbol{\pi} = -G \Delta \mathbf{W} \boldsymbol{\pi}', \quad (1)$$

where $G = \mathbf{W}^D$ is the Drazin inverse [11–13]. It satisfies $\mathbf{G} \mathbf{W} = \mathbf{W} \mathbf{G} = \mathbf{I} - \boldsymbol{\pi} \mathbf{1}^\top$, $\mathbf{G} \boldsymbol{\pi} = 0$, and $\mathbf{1}^\top \mathbf{G} = 0$. Indeed, subtracting the two stationary equations gives $\mathbf{W} \Delta \boldsymbol{\pi} = -\Delta \mathbf{W} \boldsymbol{\pi}'$. Multiplication by G and normalization give Eq. (1), without assuming that $\Delta \mathbf{W}$ is small.

The Drazin inverse underlies current-cumulant calculations [14], expansions under slow driving [15], fluctuation-response relations for state observables [16] and state-current correlations [17], responses in the Curie–Weiss model [18], and finite perturbations of Markov jump processes [19]. The relation $\tau_{j \rightarrow i} = (G_{ij} - G_{ii})/\pi_i$ connects this approach to response formulas and bounds using mean first-passage times [2, 19–21]. Alternatively, normalization gives an invertible matrix from the rate matrix [22, 23].

III. MUTUAL LINEARITY

A. State observables

We start with the derivation of mutual linearity for probabilities and state observables. For the second matrix $\mathbf{W}'$, we change W_{ij} and W_{ji} , keeping all other non-diagonal rates fixed and adjusting the diagonal elements to preserve probability conservation. Then, only components i and j of the vector $\Delta \mathbf{W} \boldsymbol{\pi}'$ can be nonzero, and they sum to zero. Equation (1) therefore gives

$$\begin{aligned} \Delta \pi_n &= (G_{nj} - G_{ni})(\Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i) \\ &\equiv h_n \alpha, \end{aligned} \quad (2)$$

where $h_n \equiv G_{nj} - G_{ni}$ is calculated from the rates before the change and does not depend on ΔW_{ij} or ΔW_{ji} . All dependence on these rate changes enters the second factor $\alpha \equiv \Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i$. Repeating Eq. (2) for state m and eliminating α gives

$$\Delta \pi_n = \frac{h_n}{h_m} \Delta \pi_m, \quad h_m \neq 0. \quad (3)$$

* timur.aslyamov@uni.lu

† massimiliano.esposito@uni.lu

Thus, (π'_m, π'_n) lies on one straight line for every choice of the new edge rates. This property was recently discovered in Ref. [5].

More generally, for state observables $O = \sum_n O_n \pi_n$ and $Q = \sum_n Q_n \pi_n$ with fixed weights, the same argument gives

$$\Delta O = \frac{\sum_n O_n h_n}{\sum_n Q_n h_n} \Delta Q, \quad (4)$$

provided that the denominator is nonzero.

Let us now consider the linear-response limit of Eq. (2):

$$\partial_{W_{ij}} \pi_n = h_n \pi_j, \quad \partial_{W_{ji}} \pi_n = -h_n \pi_i, \quad (5)$$

so that $-\partial_{W_{ij}} \pi_n / \partial_{W_{ji}} \pi_n = \pi_j / \pi_i$ whenever $h_n \neq 0$. This recovers the response ratio of Ref. [5]. Interestingly, our dynamical fluctuation-response theory [24] gives the same ratio for dynamical responses to impulsive rate perturbations at time t , with π_j / π_i replaced by $p_j(t) / p_i(t)$. Thus, this response ratio can be obtained without invoking mutual linearity under finite rate changes.

B. Currents

Now we turn to mutual linearity between the steady-state currents. The stationary current from l to k is $J_{kl} \equiv W_{kl} \pi_l - W_{lk} \pi_k$. For an edge whose transition rates W_{kl} and W_{lk} are kept fixed, Eq. (2) gives

$$\Delta J_{kl} = (W_{kl} h_l - W_{lk} h_k) \alpha, \quad (6)$$

where only the change in the stationary distribution contributes. On the perturbed edge, the rates themselves also change:

$$\begin{aligned} \Delta J_{ij} &= W_{ij} \Delta \pi_j - W_{ji} \Delta \pi_i + \Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i \\ &= (1 + W_{ij} h_j - W_{ji} h_i) \alpha. \end{aligned} \quad (7)$$

Eliminating α between the currents on (i, j) and another edge (k, l) gives

$$\Delta J_{kl} = \frac{W_{kl} h_l - W_{lk} h_k}{1 + W_{ij} h_j - W_{ji} h_i} \Delta J_{ij}, \quad (8)$$

which is the mutual linearity of stationary currents [1–4]. Both current changes contain the same α , so the coefficient relating them is independent of the new edge rates, provided the denominator in Eq. (8) is nonzero.

IV. FURTHER APPLICATIONS

We now apply the common-factor argument to Laplace-transformed dynamics and stationary complex activities. First, consider a linear system $A\mathbf{x} = \mathbf{b}$ with nonsingular A and fixed $\mathbf{b}$. If only A_{ij} changes and A' is nonsingular, subtracting the equations and eliminating $\Delta A_{ij} x'_j$ between n and m gives

$$\begin{aligned} \Delta x_n &= -(A^{-1})_{ni} \Delta A_{ij} x'_j, \\ \Delta x_n &= \frac{(A^{-1})_{ni}}{(A^{-1})_{mi}} \Delta x_m, \quad (A^{-1})_{mi} \neq 0. \end{aligned} \quad (9)$$

The same proportionality holds for fixed linear observables. Moreover, this argument applies whenever $\Delta A \mathbf{x}' = \mathbf{u} \alpha$, with fixed $\mathbf{u}$: the perturbation enters the equations in fixed proportions, with only its scalar amplitude α changing. Then $\Delta \mathbf{x} = -A^{-1} \mathbf{u} \alpha$, so eliminating α again gives mutual linearity.

A. Laplace-transformed dynamics

The frequency-domain extension follows from this argument using $A = sI - W$. For time-independent rates, the Laplace-transformed master equation is $(sI - W) \tilde{\mathbf{p}}(s) = \mathbf{p}(0)$. We keep the same initial distribution in both systems and define the resolvent matrix $R(s) = (sI - W)^{-1}$ for $\text{Re } s > 0$. Changing the rates of edge (i, j) and eliminating the common factor between states n and m results in

$$\begin{aligned} \Delta \tilde{p}_n &= h_n(s) (\Delta W_{ij} \tilde{p}'_j - \Delta W_{ji} \tilde{p}'_i), \\ \Delta \tilde{p}_n &= \frac{h_n(s)}{h_m(s)} \Delta \tilde{p}_m, \quad h_m(s) \neq 0, \end{aligned} \quad (10)$$

where $h_n(s) = R_{ni}(s) - R_{nj}(s)$ and all transformed probabilities are evaluated at the same s . This recovers the Laplace-domain probability relations [6, 7]. Repeating Eqs. (6)–(8) with $\tilde{\mathbf{p}}(s)$ and $h_n(s)$ gives the corresponding current relations [1]. These relations hold for the Laplace transforms. At a fixed time, varying the edge rates need not produce a straight-line relation between two probabilities or two currents [7].

B. Chemical reaction networks

The same argument also applies to complex activities in chemical reaction networks. We recover the relation of Ref. [10] directly through the Drazin inverse, avoiding the spanning-tree construction and the associated coplanarity argument. Here a *complex* is the combination of species on either side of a reaction. For example, $A + B \rightleftharpoons C$ connects the complexes $A + B$ and C . If x_A, x_B, x_C are the species concentrations, their *activities* are $\psi_{A+B} = x_A x_B$ and $\psi_C = x_C$. Mass-action kinetics means that each reaction rate is its rate constant times the activity of its source complex.

View the complexes as graph vertices and the reactions as edges. The matrix K plays the same role as W above: K_{ij} is the rate constant for $j \rightarrow i$, and each column sums to zero. We consider bidirectional reactions on one connected graph (one linkage class), with zero deficiency [25]. This last structural condition ensures that every positive stationary state is *complex balanced*: at each complex, the total incoming reaction flux equals the total outgoing flux. Thus $K\boldsymbol{\psi} = 0$. This equation is linear in the activities, even though the activities can depend nonlinearly on concentrations. Unlike probabilities, they are not normalized: their sum $S = \sum_n \psi_n$ can change with the rates.

Now vary only the forward and backward rate constants r_+ and r_- of one reaction $j \rightleftharpoons i$. Remove that reaction in both directions to define a reference matrix K_0 , and assume

that the remaining graph is connected (the removed edge is not a bridge). Then K_0 depends only on the fixed rates and has a normalized stationary vector π^0 and Drazin inverse G_0 . The removed reaction carries the net current $J = r_+ \psi_j - r_- \psi_i$: it supplies complex i and drains complex j . The remaining reactions must balance this current, so $K_0 \psi = -(\mathbf{e}_i - \mathbf{e}_j)J$, where $\mathbf{e}_i$ is a unit vector. Using $G_0 K_0 = I - \pi^0 \mathbf{1}^\top$ gives

$$\psi_n = S\pi_n^0 + h_n J, \quad h_n = (G_0)_{nj} - (G_0)_{ni}.$$

All dependence on the varied rates therefore enters through two scalars, S and J ; the vectors π^0 and $\mathbf{h}$ stay fixed. Writing this expression for three complexes a, b, c and eliminating both scalars gives

$$C_{bc}\psi_a + C_{ca}\psi_b + C_{ab}\psi_c = 0, \quad (11)$$

where $C_{ab} = \pi_a^0 h_b - \pi_b^0 h_a$ depends only on the fixed rates. If $C_{bc} \neq 0$, two stationary activities determine the third for

every choice of r_+ and r_- . The relation concerns activities, such as $x_A x_B$, and does not in general imply a linear relation between individual species concentrations. For probabilities, normalization fixes $S = 1$, so only J needs to be eliminated and two components suffice, recovering the earlier affine relation.

V. CONCLUSION

The classical finite-response identity yields mutual linearity whenever the perturbation source has a fixed direction. This gives the stationary and Laplace-space relations without expanding in the rate changes. For the complex-balanced chemical networks considered here, the activity scale supplies a second amplitude; eliminating both gives the homogeneous three-activity relation.

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